Assignment 4

Coverage & Interface

Required:

1) To re-structure the implemented test environment for UART-TX in last assignment to be connectivity interface-based, implement interface to make your connections.

2) Implement covergroup with comprehensive coverpoints that covers the corner cases with the stimulus for
   - Coverpoint for covering all combinations of parallel data_in.
   - Coverpoint has bins for corner values only for data_in like all ones, all zeros, sequence from all ones to all zeros and vice-versa.
   - Coverpoint for reset with bins for its value 1 and for its value of 0.
   - Coverpoint for tx_start with appropriate bins.
   - Coverpoint with different indicative bins for all types of parity.

3) Generate the functional and Code coverage reports and it is **mandatory** to reach 100% coverage. Not hitting this percentage will require u to enhance your stimulus in test environment.

U will need to add snippets for coverage as mentioned below

Deliverables:

U have to deliver

- Zip Folder contains:
  - The implemented **.sv** files and the **design files** and **run.do file**.

- **Separated PDF** contains:
  - Snippets for the implemented code.
  - Snippet for the waveform shows the different testcases.
  - Snippets for the logs show the all printed values.
  - Snippets for functional coverage report.
  - Snippets for code coverage report for all parameters [Line / toggle / branch /condition/…]
  - Snippet for Do file you have used for automating the process.

The delivered zip file must be named like **your_name_Assignment_1.rar** for example:
Hassan_Khaled_Assignment_4.rar also the PDF file Hassan_Khaled_Assignment_4.pdf

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Good Luck

Hassan Khaled